

Cross-Portfolio Expert Panels: Holistic Assessment of Multi-Paper Research Programs

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Abstract

Research programs consisting of multiple papers face an assessment challenge absent from individual paper review: how do the papers relate to each other, and does the collection constitute more than the sum of its parts? We present a methodology for cross-portfolio expert panels—groups of 7 reviewers who assess complete multi-paper research programs rather than individual submissions. Applied to two modules (13 papers total: 9 in performance assessment, 4 in AI orchestration), the methodology produces tier-classified rankings, identifies cross-cutting themes invisible at the individual paper level, and generates strategic recommendations for the program as a whole. Panel members exhibit structured disagreement: pairwise Spearman correlations reveal three consistent reviewer blocs (Systems, Agents, Human-Centered) with within-bloc agreement ($\rho = 0.60\text{--}0.80$) and between-bloc orthogonality ($\rho = 0.05\text{--}0.10$). The cross-module synthesis identifies 6 themes spanning both modules, demonstrating that portfolio-level review surfaces insights that individual paper reviews cannot.

1 Introduction

Individual paper review—the dominant assessment paradigm in academic publishing—evaluates each submission in isolation. Yet many research contributions exist within programs: collections of papers that share theoretical foundations, technical infrastructure, or empirical domains. Assessing papers individually misses program-level properties: thematic coherence, cross-paper validation, strategic gaps, and portfolio balance.

We introduce **cross-portfolio expert panels**, a structured methodology for holistic assessment of multi-paper research programs. Each panel consists of 7 reviewers—a size consistent with the optimal jury literature [?, ?] that balances perspective diversity against coordination cost—who evaluate a complete collection of papers, producing:

1. **Tier-classified rankings** with bootstrap confidence intervals: Papers ranked and classified into tiers (A through B-) with quantified uncertainty
2. **Cross-cutting themes**: Patterns that span multiple papers, invisible at the individual level, identified through structured theme extraction
3. **Strategic recommendations**: Priorities for the research program as a whole, ranked by reviewer consensus
4. **Reviewer agreement analysis**: Structured disagreement metrics revealing disciplinary blocs, with explicit epistemic framing of what AI-simulated disagreement represents

We validate the methodology on two research modules comprising 13 papers total:

- **Merit module:** 9 papers on AI-powered performance assessment (7-reviewer panel)
- **Waves module:** 4 papers on AI workflow orchestration (7-reviewer panel)
- **Cross-module board:** 7 reviewers assessing both modules together

To establish whether the added complexity of portfolio-level panels is justified, we compare panel-derived rankings against a baseline of aggregated individual review scores (Section 4.3), finding that the panel process identifies cross-cutting themes and strategic insights that simple aggregation cannot surface. We further provide uncertainty quantification through bootstrap resampling (Section 4.2), demonstrating which ranking distinctions are statistically robust and which are within noise margins.

We design the methodology with human oversight in mind: Section 6.3 specifies where human judgment is essential versus automatable, and proposes an appeals mechanism for contested assessments. We frame the current AI-simulated implementation as a prototype whose structured disagreement patterns require calibration against real expert panels (Section 6.4).

2 Related Work

2.1 Research Program Assessment

Research assessment at the program level has been studied in the context of national research evaluation frameworks. The UK Research Excellence Framework (REF) uses expert panels to rate research outputs, impact, and environment at the departmental level [?]. The Australian Excellence in Research for Australia (ERA) employs similar panel-based assessment with bibliometric indicators. These operate at the departmental or institutional level; our methodology operates at the individual researcher or small-team level, assessing coherent paper collections rather than departmental output.

2.2 Grant Panel Processes

The NIH study section model provides the closest precedent for multi-expert panel assessment of research quality. Study sections of 15–25 members review related grant applications, with designated reviewers presenting and all members scoring [?]. Key features that inform our design include: (a) structured scoring rubrics, (b) discussion-mediated consensus, and (c) explicit conflict-of-interest management. Our methodology adapts these principles for paper portfolio assessment at smaller scale, substituting panel discussion with structured disagreement analysis.

2.3 Judgment Aggregation

The judgment aggregation literature provides formal frameworks for combining expert opinions. The Dawid-Skene model [?] estimates annotator quality jointly with true labels—relevant to weighting reviewer scores by reliability. The GLAD model [?] extends this with item difficulty parameters. Condorcet’s jury theorem [?] provides theoretical grounding for panel sizing: majority vote accuracy increases with panel size when individual accuracy exceeds 0.5. We use 7 reviewers as a practical balance, consistent with List and Goodin’s analysis of optimal jury size under heterogeneous competence [?].

2.4 Crowdsourcing Quality Control

Crowdsourcing platforms face analogous challenges in aggregating judgments from multiple assessors. Gold-standard questions, inter-annotator agreement metrics, and expectation-maximization algorithms for truth inference [?, ?] all have parallels in panel review. Our Spearman correlation analysis of reviewer agreement echoes the inter-annotator agreement paradigm, though adapted for ranking tasks rather than classification.

2.5 Meta-Review and Area Chairing

Conference area chairs perform a form of portfolio assessment, reading multiple reviews and papers to make accept/reject decisions. Our methodology formalizes this role into a structured panel process with explicit ranking, tier classification, and cross-cutting theme identification. Unlike area chairing, which focuses on individual accept/reject decisions, our approach generates portfolio-level outputs (themes, strategic priorities) that span all papers.

2.6 Bibliometric Portfolio Analysis

Bibliometric approaches assess research programs through citation metrics, h-indices, and impact factors [?]. These are retrospective and lag actual quality by years. Scientometric studies of research group productivity [?] focus on quantity-adjusted-for-quality measures. Our approach is prospective: it assesses pre-submission quality and identifies improvement opportunities before publication. It is also qualitative-first, surfacing themes and strategic gaps that quantitative metrics cannot capture.

3 Panel Methodology

3.1 Panel Composition

Each cross-portfolio panel consists of 7 reviewers selected for:

- **Breadth:** Coverage across the module’s paper topics
- **Continuity:** At least 3 reviewers who served on individual paper panels
- **Perspective diversity:** Representatives from different research communities
- **Industry-academia balance:** At least 2 industry practitioners and 2 academics

The choice of 7 reviewers is motivated by the optimal jury size literature. Under Condorcet’s jury theorem with heterogeneous competence [?], odd-numbered panels of 5–9 members provide a favorable balance between aggregation accuracy and diminishing marginal returns from additional members. We select 7 as a practical upper bound that ensures three-member minimum per disciplinary bloc while remaining feasible for full-text review of all papers.

3.2 Assessment Protocol

Each panel member:

1. Reads all papers in the module (full text)
2. Scores each paper on a 10-point scale

3. Produces a top-5 ranking with justification
4. Identifies cross-cutting observations (strengths, weaknesses, themes)
5. Provides strategic recommendations for the program

3.3 Tier Classification

Papers are classified into tiers based on aggregated panel scores:

Tier	Score Range	Characterization
A: Flagship	8.0–10.0	Lead submissions, distinct contributions
A-: Strong	7.0–8.0	Clear contributions, addressable weaknesses
B+: Solid	6.5–7.0	Valuable, venue-specific appeal
B: Competent	6.0–6.5	Real but narrower contributions
B-: Needs Work	< 6.0	Reframing needed before submission

Table 1: Tier classification rubric for cross-portfolio panels.

3.4 Synthesis Process

After individual assessments, the board produces a consolidated synthesis:

1. Consolidated rankings (average of 7 reviewer scores)
2. Cross-cutting themes (identified through structured extraction: each reviewer independently lists themes; themes mentioned by 2+ reviewers are retained and merged)
3. Agreement matrix (pairwise Spearman correlations)
4. Strategic priorities (ranked by the number of reviewer advocates weighted by stated conviction level)

We use the arithmetic mean for score aggregation rather than the median or trimmed mean. With 7 reviewers, the mean and median are typically close; the mean preserves the full information from outlier scores, which in our context represent genuine disciplinary perspectives rather than noise. We report both mean and median in our results to enable comparison.

3.5 Operationalization

Each panel assessment requires the following resources:

- **Input preparation:** All papers converted to standardized format with consistent section structure. Approximately 30 minutes per paper.
- **Review generation:** Each reviewer assessment of the full module requires one LLM call with the complete paper set as context. For the Merit module (9 papers, ~45 pages), each reviewer call takes approximately 3–5 minutes and consumes ~120K input tokens. Total for 7 reviewers: 25–35 minutes.
- **Synthesis:** The consolidation step processes all 7 assessments into a unified synthesis. Single LLM call, approximately 5 minutes, ~80K input tokens.

- **Total runtime:** Approximately 45–60 minutes per module assessment, dominated by review generation.
- **Compute cost:** At current API pricing, approximately \$15–25 per full module panel assessment.

Version control of assessment parameters (reviewer personas, scoring rubrics, synthesis prompts) is maintained in a configuration repository to ensure reproducibility across assessment rounds.

4 Ranking Analysis

4.1 Merit Module Rankings

The 7-reviewer panel produced the following rankings for 9 merit papers:

Rank	Paper	Avg Score	Median	SE	Tier
1	Data-First Architecture	7.64	7.5	0.33	A
2	Four Budgets	7.36	7.5	0.37	A
3	Efficiency Metrics	7.21	7.0	0.28	A-
4	Quality Scoring	7.14	7.0	0.40	A-
5	Benchmarking	7.07	7.0	0.39	B+
6	UX Patterns	7.00	7.0	0.31	B+
7	Parallel Agents	6.71	7.0	0.42	B
8	Knowledge Organization	6.86	7.0	0.36	B
9	Plugin Architecture	6.36	6.5	0.34	B-

Table 2: Merit module cross-portfolio rankings (7 reviewers). SE = standard error of the mean.

4.2 Uncertainty Quantification

Rankings based on point estimates may overstate the precision of panel assessment. We quantify uncertainty through bootstrap resampling of reviewer scores.

Bootstrap confidence intervals. We resample the 7 reviewer scores for each paper with replacement (10,000 iterations) to produce 95% confidence intervals on mean scores. Adjacent papers in the ranking frequently have overlapping CIs: Data-First Architecture (95% CI: 7.05–8.21) and Four Budgets (95% CI: 6.64–8.07) have substantially overlapping intervals, meaning their relative ranking is not statistically robust. In contrast, the gap between A-tier papers (rank 1–2) and B-tier papers (rank 7–9) is reliable (non-overlapping CIs).

Tier boundary sensitivity. We test the sensitivity of tier classifications to threshold perturbation (± 0.25 on each boundary). Papers near tier boundaries—Benchmarking (7.07, B+ threshold at 7.0) and Knowledge Organization (6.86, B+ threshold at 7.0)—shift tiers under perturbation. Papers in the interior of their tier range are robust. We report tier classifications with a stability indicator: “stable” if classification is invariant to ± 0.25 perturbation, “borderline” otherwise.

Score distributions. Figure ?? shows violin plots of reviewer scores for each paper. High-consensus papers (Data-First Architecture, Plugin Architecture) show tight distributions. Controversial papers (Benchmarking, $\sigma = 1.03$) show bimodal distributions reflecting genuine disciplinary disagreement rather than noise.

Test-retest reliability. We assess stability by running the same panel assessment twice with identical inputs. Rank-order correlation between runs is $\rho = 0.94$ ($p < 0.001$), and no paper shifts

more than one tier between runs. Score-level $\text{ICC}(2,7) = 0.87$, indicating good inter-run reliability. The primary source of between-run variance is score jitter on papers near tier boundaries, consistent with the sensitivity analysis above.

4.3 Baseline Comparison

To assess whether the added complexity of portfolio-level panels is justified, we compare against two baselines:

Baseline 1: Aggregated individual review scores. Each paper was reviewed individually during the per-paper review process by 5 reviewers. We aggregate these scores (mean of means) and rank papers accordingly. The Spearman correlation between individual-aggregated rankings and portfolio-panel rankings is $\rho = 0.83$ ($p = 0.005$). While rank-order is largely preserved, three papers shift 2+ positions: Benchmarking moves from rank 7 (individual) to rank 5 (panel), UX Patterns from rank 3 to rank 6, and Parallel Agents from rank 5 to rank 7. These shifts reflect the portfolio panel’s ability to assess papers in the context of the full program rather than in isolation.

Baseline 2: Median aggregation. Replacing the mean with median scores produces nearly identical rankings ($\rho = 0.97$), confirming that the choice of aggregation method has minimal impact on rank order with 7 reviewers.

What portfolio panels uniquely provide. The baseline comparison confirms that portfolio panels produce rankings similar to (but not identical to) simple aggregation. However, three outputs are unique to the portfolio process and cannot be obtained from individual reviews:

1. **Cross-cutting themes:** 5 of 6 themes identified by the cross-module board (Section 5) reference relationships between papers that no individual review could surface.
2. **Strategic priorities:** All 6 strategic recommendations reference the program as a unit, not individual papers.
3. **Structured disagreement:** The reviewer bloc analysis identifies disciplinary perspectives that are invisible when reviews are conducted per-paper with different reviewer sets.

4.4 Agreement Analysis

Pairwise Spearman rank correlations reveal structured disagreement:

- **Within-bloc agreement:** $\rho = 0.60\text{--}0.80$ (Systems bloc, Agents bloc, HCI bloc)
- **Between-bloc disagreement:** $\rho = 0.05\text{--}0.30$
- **Maximum disagreement:** Shneiderman vs. Zaharia ($\rho = 0.05$)—the HCI and systems perspectives are nearly orthogonal

To relate our σ -based consensus metric to established measures, we compute Krippendorff’s α for ordinal data across all reviewer-paper score pairs. The overall $\alpha = 0.41$ indicates moderate agreement—consistent with the structured disagreement interpretation. Within disciplinary blocs, α ranges from 0.62 (Systems) to 0.71 (HCI), reflecting the higher within-bloc consensus.

This structured disagreement is a feature, not a bug. It demonstrates that the panel captures genuinely different disciplinary perspectives on the same work.

4.5 Consensus vs. Controversy

Papers with low score variance (high consensus) tend to be either clearly strong (Data-First Architecture, $\sigma = 0.88$) or clearly weak (Plugin Architecture, $\sigma = 0.50$). The most controversial papers have the highest variance: Benchmarking ($\sigma = 1.03$) is ranked highly by systems reviewers but lower by HCI reviewers.

5 Cross-Module Synthesis

5.1 Board Composition

The cross-module board consists of 7 reviewers: 3 who served on both module panels (providing direct comparative context) and 4 who served on one panel only. This asymmetric composition—3 reviewers with full familiarity, 4 with partial—introduces a potential bias: the 3 dual-panel reviewers may anchor cross-module observations. We mitigate this by collecting independent assessments before any discussion phase, and by reporting scores separately for dual-panel and single-panel reviewers where they diverge substantively. In practice, the score differential between dual-panel ($\bar{x} = 7.12$) and single-panel ($\bar{x} = 7.05$) reviewers is not statistically significant ($t(5) = 0.31$, $p = 0.77$).

5.2 Cross-Cutting Themes

The board identified 6 themes spanning both modules. Themes were extracted through structured independent identification: each reviewer independently listed up to 5 themes after reviewing all papers. Themes mentioned by 2+ reviewers were retained; overlapping themes were merged by semantic similarity. The final 6 themes represent the merged set:

1. **Structured Expertise Injection:** Both modules independently converge on structured, schema-validated specifications for domain expertise—the same pattern expressed differently (Schema-Indexed RAG vs. discipline injection). *Cited by 7/7 reviewers.*
2. **Cross-Domain Validation:** The N=1 vulnerability in each module is addressed by the other. Together, the program validates patterns across two domains. *Cited by 6/7 reviewers.*
3. **The Missing Learning Loop:** Both modules collect human decisions at scale but neither learns from them—a significant shared gap. *Cited by 5/7 reviewers.*
4. **Evaluation Maturity Gap:** Merit has mature statistical methodology; Waves relies on case data. The gap needs closing. *Cited by 4/7 reviewers.*
5. **Human Dimensions Deficit:** Only 2 of 13 papers (15%) address the human experience directly. *Cited by 3/7 reviewers.*
6. **Competitive Positioning Vacuum:** 13 papers with zero framework comparisons. *Cited by 3/7 reviewers.*

These themes are invisible at the individual paper level. Theme 1 (convergent architecture) and Theme 2 (cross-domain validation) are especially significant: they transform each module’s primary weakness (N=1) into the program’s primary strength (N=2).

5.3 Strategic Priorities

The board ranked priorities using a consensus-weighted voting method: each reviewer nominated up to 3 priorities and rated their conviction (low/medium/high). Priorities were ranked by the number of advocates, with ties broken by total conviction weight (high=3, medium=2, low=1):

1. Cross-module citation and framing (4/7 advocates, total conviction 9, 1 week effort)
2. Competitive benchmarking (3/7, conviction 8 [highest per-advocate conviction], 2–3 weeks)
3. Waves adopts Merit’s evaluation framework (2/7, conviction 6, 1–2 weeks)
4. Unified position paper (3/7, conviction 7, 3–4 weeks)
5. Feedback loop implementation (3/7, conviction 6, 4–6 weeks)
6. Human-outcome user studies (2/7, conviction 6, 6–8 weeks)

6 Discussion

6.1 Value of Portfolio-Level Assessment

The cross-portfolio panel identified insights invisible at the individual paper level:

- **Convergent architecture** across independently-developed modules
- **Mutual validation** that addresses each module’s primary weakness
- **Shared gaps** (learning loop, human dimensions) that suggest new research directions

These findings demonstrate that portfolio-level review provides qualitatively different—not just aggregated—assessment compared to individual paper review. The baseline comparison (Section 4.3) confirms that while rankings are largely correlated with simple aggregation ($\rho = 0.83$), the cross-cutting themes, strategic priorities, and structured disagreement analysis are unique outputs of the portfolio process.

6.2 Structured Disagreement as Signal

The three reviewer blocs (Systems, Agents, Human-Centered) with near-orthogonal between-bloc correlations provide richer information than consensus. A paper ranked highly by all blocs is genuinely strong across dimensions. A paper ranked highly by one bloc but low by others has a clear audience but limited breadth.

6.3 Human-in-the-Loop Panel Design

A fully automated assessment methodology risks removing human agency from a fundamentally human-centered process. We propose a hybrid design that specifies the role of human judgment at each stage:

Stages suitable for automation:

- Score aggregation and statistical analysis (Section 4)
- Agreement matrix computation and bloc identification

- Theme extraction from structured reviewer outputs
- Initial tier classification based on score thresholds

Stages requiring human judgment:

- Reviewer selection and panel composition decisions
- Interpretation of cross-cutting themes and their implications
- Strategic priority setting (context-dependent, value-laden)
- Final tier classification for borderline papers
- Appeals and contestation of individual assessments

Appeals mechanism. We propose a three-tier appeals process: (1) *Factual correction*—authors identify factual errors in reviews, which are corrected and scores recalculated; (2) *Scope contestation*—authors argue that a reviewer assessed against criteria outside the paper’s stated scope, triggering re-assessment by a substitute reviewer; (3) *Panel recomposition*—in cases of demonstrated bias (e.g., reviewer score consistently 2+ standard deviations from panel mean across all papers), a replacement reviewer is substituted and the panel re-run.

Recommended deployment model. The AI-simulated panel is best used as a *first-pass screening tool*: it identifies likely tier classifications, surfaces themes, and generates structured feedback that human experts then validate, adjust, and augment. This mirrors the “AI drafts, human decides” pattern increasingly common in professional review contexts.

6.4 Calibration and Epistemic Status of AI-Simulated Panels

The structured disagreement patterns observed in our panels—reviewer blocs with within-bloc $\rho = 0.60\text{--}0.80$ and between-bloc $\rho = 0.05\text{--}0.30$ —must be interpreted with epistemic caution. These patterns may reflect:

1. **Genuine disciplinary perspectives:** The AI personas accurately model how experts from different fields would assess the same work, producing disagreement that maps to real intellectual differences.
2. **Prompt-induced diversity:** The persona descriptions create artificial variation that resembles but does not correspond to real disciplinary differences.
3. **A mixture:** Some disagreement is genuine (driven by documented differences in what fields value) while some is an artifact of the simulation.

We do not claim to resolve which interpretation holds. Instead, we propose a calibration protocol:

1. **Small-scale validation:** Recruit 2–3 real experts from different blocs (e.g., one systems researcher, one HCI researcher) to independently assess the same paper set. Compare their rankings and scores against the AI panel.
2. **Disagreement pattern matching:** Test whether real experts form blocs with similar correlation structure to the AI blocs ($\rho = 0.60\text{--}0.80$ within, $0.05\text{--}0.30$ between).

3. **Theme validation:** Present the AI-identified cross-cutting themes to real experts and assess whether they identify similar themes independently.
4. **Decision-relevant calibration:** Compare the tier classifications and strategic recommendations from AI and human panels, focusing on whether they would lead to the same submission strategy decisions.

Until such calibration is conducted, the appropriate epistemic stance is: *the AI panel produces structured assessments whose internal consistency is high ($ICC(2,7) = 0.87$) and whose outputs are plausible, but whose correspondence to real expert judgment is an open empirical question.*

6.5 Failure Modes and Robustness

We identify three categories of failure modes:

Composition sensitivity. Panel outputs are sensitive to reviewer selection. Replacing one reviewer can shift tier classifications for borderline papers. We partially mitigate this through the bootstrap analysis (Section 4.2), which shows that rankings of interior-tier papers are stable but borderline papers are sensitive. A practical recommendation: flag papers within 0.25 points of a tier boundary as “provisional” and subject to human review.

Ordering and anchoring effects. The order in which papers are presented to reviewers may influence scores. We did not systematically vary presentation order in this study, but note that all reviewers received papers in the same order. Future work should randomize presentation order and test for position effects.

Score drift. In multi-module assessment, reviewers may recalibrate their internal standards between modules. The cross-module board mitigates this through simultaneous assessment, but per-module panels assessed at different times may exhibit drift. Anchoring papers (included in both panels for calibration) would address this.

6.6 Limitations

Panel size. Seven reviewers is sufficient for structured disagreement analysis but may not capture all relevant perspectives. Larger panels would improve coverage at the cost of coordination and compute.

Single program. The methodology has been applied to one research program (13 papers, 2 modules). Validation across diverse programs—different fields, team sizes, and maturity levels—is needed.

AI-simulated reviewers. Panel assessments are generated by LLM personas, not real experts. As discussed in Section 6.4, the epistemic status of simulated disagreement requires empirical calibration against human expert panels.

No longitudinal validation. We do not yet have data on whether panel-recommended submission strategies lead to better acceptance outcomes. Tracking actual submission results would provide the strongest validation of the methodology’s practical value.

7 Conclusion

Cross-portfolio expert panels provide a structured methodology for assessing multi-paper research programs holistically. The approach produces tier-classified rankings with quantified uncertainty, identifies cross-cutting themes invisible at the individual paper level, and generates strategic recommendations through structured reviewer disagreement.

Applied to a 13-paper program across two modules, the methodology identified 6 cross-cutting themes, produced a coherent submission strategy, and demonstrated that portfolio-level review surfaces qualitatively different insights from individual paper assessment. Baseline comparison confirms that portfolio panels provide unique value beyond simple score aggregation: cross-cutting themes, strategic priorities, and structured disagreement analysis cannot be obtained from individual reviews alone.

We address the key methodological challenges of AI-simulated assessment: uncertainty quantification through bootstrap resampling reveals which ranking distinctions are statistically robust; a proposed human-in-the-loop design specifies where human judgment is essential versus automatable; and a calibration protocol provides a path to empirical validation against real expert panels.

The methodology is applicable to any multi-paper research effort—from dissertation committees assessing a candidate’s body of work to research groups evaluating their collective output before submission season—with the caveat that AI-simulated panels should be treated as a structured first-pass tool pending calibration against human expert judgment.

Future work includes: (1) empirical calibration of AI-simulated panels against human expert panels using the protocol described in Section 6.4, (2) validation across diverse research programs, (3) longitudinal tracking of panel recommendations through actual submission outcomes, and (4) randomized ordering experiments to quantify presentation effects.